

S6 Computer Science Exam Solutions

Page 2

1. Define: (4 marks)

- a) **Web server:** A computer or software that delivers web content (e.g., HTML pages, images) to clients over the internet via HTTP/HTTPS protocols.
- b) **Thread:** A lightweight process or a sequence of instructions within a program that can execute independently, sharing the same memory space as other threads in the process.
- c) **Computer graphics:** The creation, manipulation, and rendering of visual images and animations using computers, often for visualization or design.
- d) **Multimedia:** The integration of multiple forms of media, such as text, audio, video, images, and animations, in a single application or presentation.

2. Which statement best explains the role of cache memory in a computer system? (1 mark)

C

3. A friend wants to connect a digital camera to their computer. Which port would you recommend? (1 mark)

C

4. Which hardware upgrade best improves video editing performance? (1 mark)

Note: Option D is corrupted. Upgrading RAM from 256 MB to 4 GB significantly improves video editing performance by allowing faster data access and multitasking.

Answer: C

Page 3

15. Which of the following is NOT a universal gate? (1 mark)

C

16. What is the decimal equivalent of the binary number 1010? (1 mark)

$$1010_2 = 1 \cdot 2^3 + 0 \cdot 2^2 + 1 \cdot 2^1 + 0 \cdot 2^0 = 8 + 0 + 2 + 0 = 10$$

B

17. Which operation is used in a sum of products expression? (1 mark)

A

18. Which Boolean identity simplifies $A + A \cdot B$? (1 mark)

$$A + A \cdot B = A \cdot (1 + B) = A \cdot 1 = A \quad (\text{using distributive law and } 1 + B = 1)$$

A

19. Explain the difference between 1's complement and 2's complement with examples for converting -13 into binary. (3 marks)

- **1's Complement:** Invert all bits of the positive number's binary representation. For $+13 = 1101$ (4-bit):

$$-13 = \text{invert}(1101) = 0010$$

- **2's Complement:** Invert bits (1's complement) and add 1. For $+13 = 1101$:

$$1's \text{ complement} = 0010, \quad 2's \text{ complement} = 0010 + 1 = 0011$$

- **Difference:** 1's complement inverts bits, while 2's complement adds 1 to the inverted result, avoiding end-around carry issues in arithmetic.

Answer:

- 1's complement of -13: 0010
- 2's complement of -13: 0011

20. What type of control structure checks a condition before executing a block of code? (1 mark)

B

21. Which of the following is true about a one-dimensional array? (1 mark)

B

22. What is the output of an algorithm that converts 10 from base 10 to binary? (1 mark)

$10_{10} = 1010_2$ (divide by 2: $10 \div 2 = 5$ R0, $5 \div 2 = 2$ R1, $2 \div 2 = 1$ R0, $1 \div 2 = 0$ R1)

B

23. In a REPEAT...UNTIL loop, when is the condition evaluated? (1 mark)

C

24. What is the purpose of using arrays in control structures? (1 mark)

B

25. Design an algorithm and flowchart that reads the grades of 10 students, calculates the average grade, and outputs whether each student passed or failed based on a pass mark of 50. (4 marks)

Algorithm:

1. Initialize sum = 0, count = 10.
2. Create array grades[10] to store grades.
3. For i = 1 to 10:
 - (a) Input grades[i].
 - (b) Add grades[i] to sum.
 - (c) If grades[i] >= 50, output "Student i: Passed".
 - (d) Else, output "Student i: Failed".
4. Calculate average = sum / count.
5. Output average.

Flowchart:

- Start
- Initialize: sum = 0, count = 10, grades[10]
- Loop (i = 1 to 10):

- Input grades[i]
- sum = sum + grades[i]
- Decision: grades[i] >= 50?
 - * Yes: Output "Student i: Passed"
 - * No: Output "Student i: Failed"
- Calculate: average = sum / count
- Output: average
- End

26. To reference storage of a variable in main memory, two operators, namely sizeof and address of (&), may be used. Using sample code, differentiate between the two operators. (3 marks)

- **sizeof:** Returns the size (in bytes) of a variable or data type.
- **address of (&):** Returns the memory address of a variable.

Sample Code (C++):

```
#include <iostream>
using namespace std;
int main() {
    int x = 10;
    cout << "Size of x: " << sizeof(x) << " bytes" << endl; // Outputs: 4
    cout << "Address of x: " << &x << endl; // Outputs: e.g., 0x7ffee4c0
    return 0;
}
```

Difference: sizeof gives the storage size (e.g., 4 bytes for an int), while & provides the memory address (e.g., a hexadecimal address).

27. A linear list of elements in which deletion can be done from one end (front) and insertion from the other (rear) is known as

Answer: Queue

28. Using CERN Using a clear example, explain inheritance in Java. (4 marks)

Inheritance: A mechanism where a class (subclass) inherits properties and methods from another class (superclass).

Example:

```
class Vehicle {
    void move() {
        System.out.println("Vehicle is moving");
    }
}
```

```

    }
}
class Car extends Vehicle {
    void honk() {
        System.out.println("Car honks");
    }
}
public class Main {
    public static void main(String[] args) {
        Car car = new Car();
        car.move(); // Inherited from Vehicle
        car.honk(); // Defined in Car
    }
}

```

Output:

```

Vehicle is moving
Car honks

```

Explanation: The `Car` class inherits the `move` method from `Vehicle` and adds its own `honk` method.

Page 4

29. Which access specifier is used to hide data from outside the class? (1 mark)

C

30. What is function overloading in C++? (1 mark)

B

31. Which function is used to allocate memory to an object in C++? (1 mark)

C

32. a) What are benefits of Java Collection Framework? (2 marks)

1. **Reusability:** Provides ready-to-use data structures (e.g., `ArrayList`, `HashMap`).
2. **Flexibility:** Supports dynamic resizing and various data manipulation operations.
3. **Consistency:** Standardized interfaces (e.g., `List`, `Set`) ensure predictable behavior.
4. **Performance:** Optimized implementations for different use cases.

32. b) Write a Java program that performs the following operations using a linked list: (6 marks)

```
import java.util.LinkedList;
public class DistrictList {
    public static void main(String[] args) {
        // i) Create a linked list
        LinkedList<String> district = new LinkedList<>();

        // ii) Add districts
        district.add("Gakenke");
        district.add("Rubawa");
        district.add("Gasabo");
        district.add("Nyagatare");
        district.add("Nyabihu");

        // iii) Add Bugesera to the front
        district.addFirst("Bugesera");

        // iv) Insert Karongi at index 3
        district.add(3, "Karongi");

        // v) Replace Gasabo with Nvarugenge
        int index = district.indexOf("Gasabo");
        if (index != -1) {
            district.set(index, "Nvarugenge");
        }

        // vi) Retrieve and display the second district
        System.out.println("Second district: " + district.get(1));

        // vii) Display the size of the list
        System.out.println("Size of list: " + district.size());

        // viii) Remove the last district
        district.removeLast();

        // Display final list
        System.out.println("Final list: " + district);
    }
}
```

Output:

Second district: Gakenke

Size of list: 7

Final list: [Bugesera, Gakenke, Rubawa, Karongi, Nvarugenge, Nyagatare, Nyabihu]

33. Which keyword is used to make a function a friend of a class? (1 mark)

A

34. What happens in External Fragmentation? (1 mark)

B

35. Which of the following is NOT a core component of Apache Tomcat? (1 mark)

D

36. What is the first step in a Servlet's life cycle? (1 mark)

D

37. What does the status code 500 indicate? (1 mark)

C

38. a) List 6 properties of a file. (3 marks)

1. Name
2. Size
3. Type (extension)
4. Creation date
5. Last modified date
6. Permissions (read/write/execute)

38. b) Give 4 examples of networking devices. (2 marks)

1. Router
2. Switch
3. Hub
4. Modem

39. Which property ensures that data is delivered to the correct recipient? (1 mark)

B

Page 5

40. a) The ability to change the logical schema without altering external schemas is called... (2 marks)

Answer: Data Independence

40. b) The constraint used to link records between two tables is a... (2 marks)

Answer: Foreign Key

41. Write a Java program to demonstrate multilevel inheritance. (5 marks)

```
class Animal {
    void eat() {
        System.out.println("Animal is eating");
    }
}
class Dog extends Animal {
    void bark() {
        System.out.println("Dog is barking");
    }
}
class BabyDog extends Dog {
    void weep() {
        System.out.println("BabyDog is weeping");
    }
}
public class Test {
    public static void main(String[] args) {
        BabyDog babyDog = new BabyDog();
        babyDog.eat();    // From Animal
        babyDog.bark();   // From Dog
        babyDog.weep();   // From BabyDog
    }
}
```

Output:

```
Animal is eating
Dog is barking
BabyDog is weeping
```

42. Authorization is the process of: (1 mark)

C

43. Explain the role of each of the following protocols: (3 marks)

- a) **DHCP**: Automatically assigns IP addresses and network configuration to devices.
- b) **ARP**: Maps IP addresses to MAC addresses for local network communication.
- c) **ICMP**: Facilitates error reporting and diagnostics (e.g., ping, traceroute).

44. Choose the suitable word(s) to fill the following blanks: (5 marks)

- a. Process
- b. Long term scheduler
- c. PCB (Process Control Block)
- d. Dispatcher
- e. Short-term scheduler

45. Assume that a database is connected to a Visual Basic form using the ADODC control. Provide the methods or actions required to: (5 marks)

- a) Add a new record: `Adodc1.Recordset.AddNew`
- b) Move to the next record: `Adodc1.Recordset.MoveNext`
- c) Move to the previous record: `Adodc1.Recordset.MovePrevious`
- d) Move to the first record: `Adodc1.Recordset.MoveFirst`
- e) Move to the last record: `Adodc1.Recordset.MoveLast`
- f) Delete the current record: `Adodc1.Recordset.Delete`

Page 6

46. Provide HTML and CSS code of the below form: (5 marks)

Note: The form description is corrupted. Below is a sample form with common fields (name, email, submit button).

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Sample Form</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      display: flex;
      justify-content: center;
```

```

        align-items: center;
        height: 100vh;
        margin: 0;
        background-color: #f0f0f0;
    }
    .form-container {
        background-color: white;
        padding: 20px;
        border-radius: 8px;
        box-shadow: 0 0 10px rgba(0,0,0,0.1);
    }
    label {
        display: block;
        margin-bottom: 5px;
    }
    input[type="text"], input[type="email"] {
        width: 100%;
        padding: 8px;
        margin: 10px 0;
        border: 1px solid #ccc;
        border-radius: 4px;
    }
    input[type="submit"] {
        background-color: #4CAF50;
        color: white;
        padding: 10px 15px;
        border: none;
        border-radius: 4px;
        cursor: pointer;
    }
    input[type="submit"]:hover {
        background-color: #45a049;
    }
</style>
</head>
<body>
    <div class="form-container">
        <form>
            <label for="name">Name:</label>
            <input type="text" id="name" name="name" required>
            <label for="email">Email:</label>
            <input type="email" id="email" name="email" required>
            <input type="submit" value="Submit">
        </form>
    </div>
</body>
</html>

```

Note: Please provide specific form details for an accurate solution.